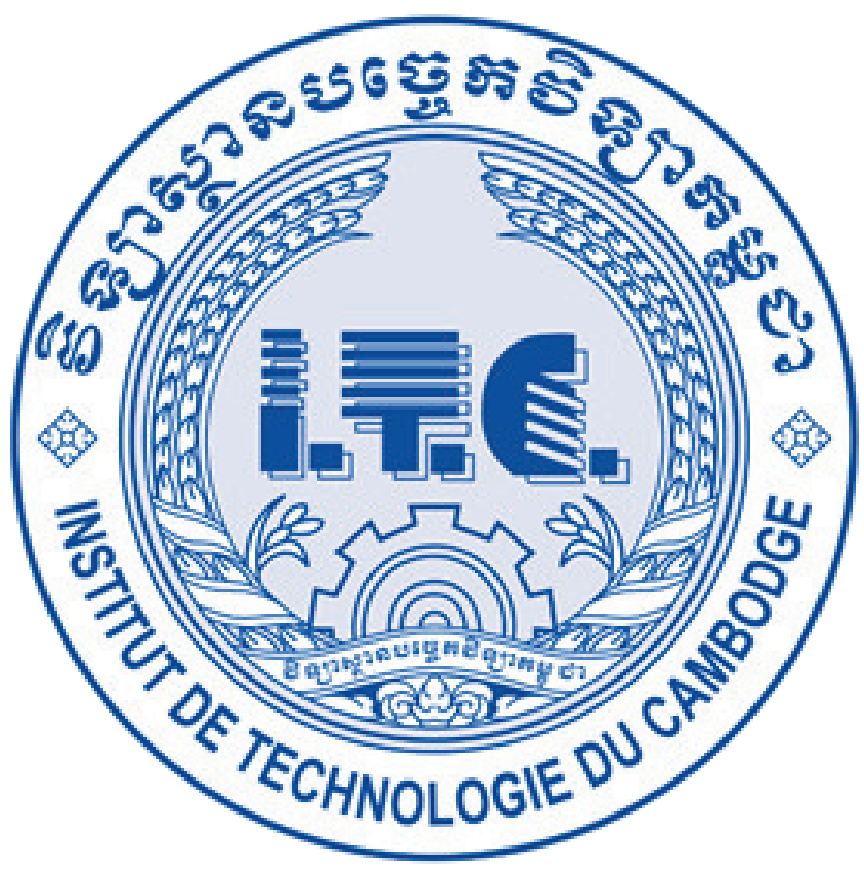




Predictive Analysis of Traffic: A Comparative Study of SARIMA, and XGBoost

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INTRODUCTION

Research Background

Traffic congestion happens when too many vehicles use the road at the same time, causing delays, pollution, and fuel waste. It’s a major issue in growing cities, especially during rush hours. Causes include rapid urbanization, limited public transport, and poor traffic management. Reducing congestion requires smart planning and accurate traffic prediction.

PROBLEM STATEMENT

With rapid economic growth, technological advancement, and urbanization, traffic congestion has become a major issue in cities. **Traffic forecasting**, which **analyzes historical** and **real-time** data to predict traffic flow, is increasingly important for improving traffic management, reducing congestion, and providing timely navigation information.

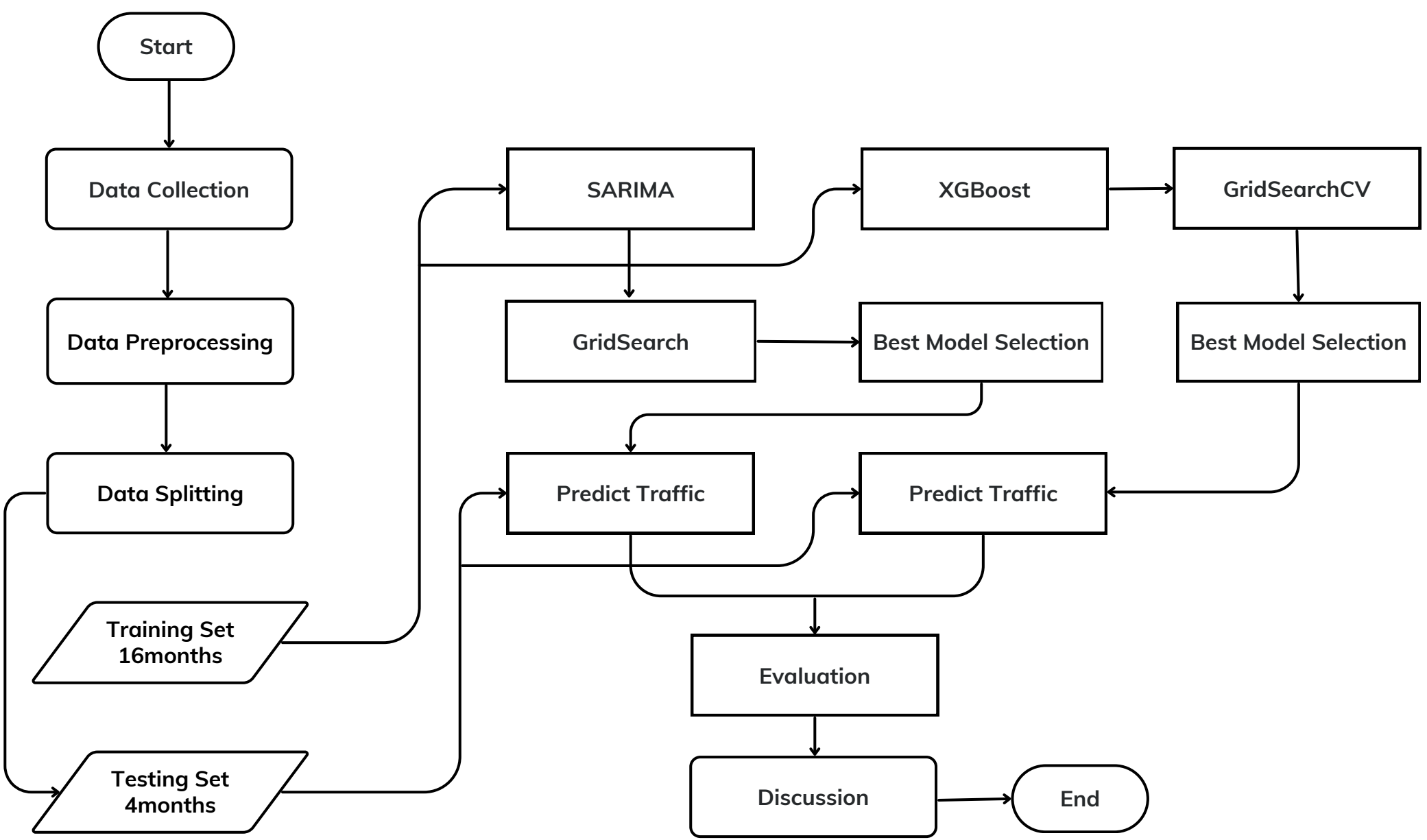


PURPOSE OF THE RESEARCH

This research aims to improve traffic prediction accuracy by using **SARIMA** to model seasonal patterns ,and **XGBoost** to capture complex, non-linear relationships in traffic data. By combining statistical and machine learning approaches, the study supports better traffic management and urban mobility planning.

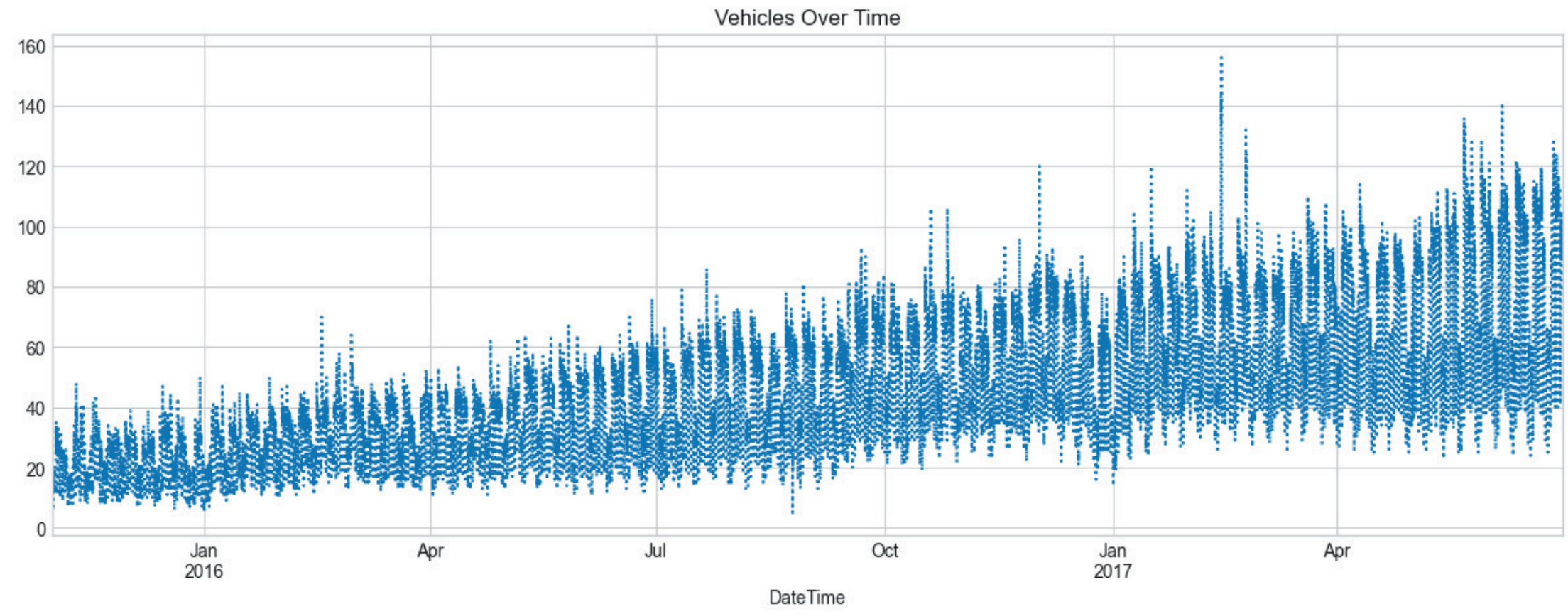
METHODOLOGY

RESEARCH FLOW



DATASET

This dataset, titled "**Traffic Prediction Dataset**", contains traffic data collected hourly from one junction within an urban area in Washington D.C. The data spans from November 1, 2015, to June 30, 2017, providing a substantial historical record to analyze urban mobility patterns.



SARIMA MODEL

The **Seasonal Autoregressive Integrated Moving Average (SARIMA)** model is a time series method designed to capture both short-term and seasonal dependencies in data. It extends the ARIMA model by adding **seasonal autoregressive (AR)** and **moving average (MA)** terms. The model is expressed as SARIMA(p, d, q)(P, D, Q)_s, where p and q are the non-seasonal orders, d is the non-seasonal differencing order, P and Q are the seasonal orders, D is the seasonal differencing order, and s is the seasonal period.

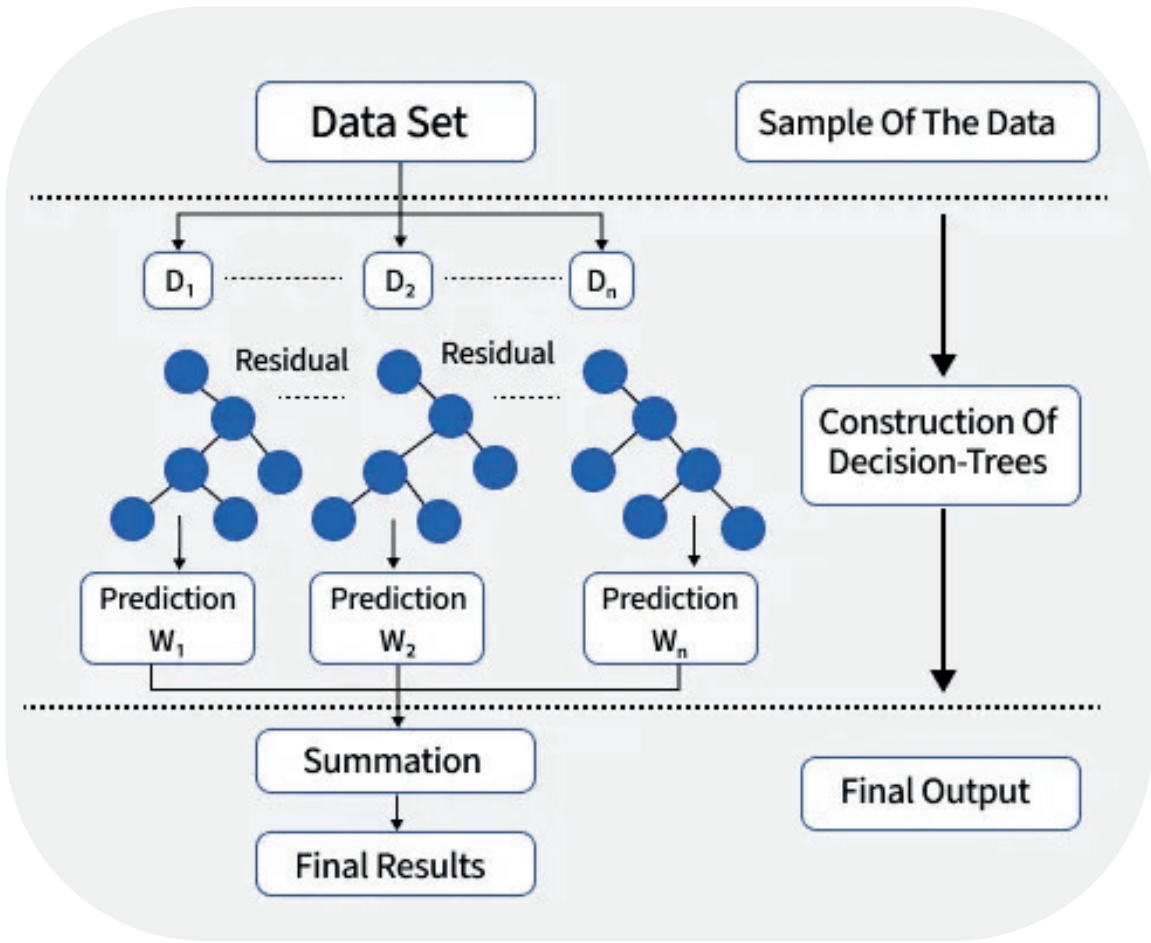
The general formula for SARIMA is:

$$(1 - \phi_1 B) (1 - \Phi_1 B^4) (1 - B) (1 - B^4) y_t = (1 + \theta_1 B) (1 + \Theta_1 B^4) e_t.$$

This model is suitable for **stationary data** with repeating seasonal patterns and is commonly used in applications like traffic or demand forecasting.

XGBOOST FOR REGRESSION

XGBoost (Extreme Gradient Boosting) is a powerful and scalable supervised learning algorithm that builds an ensemble of weak learners, typically decision trees, using gradient boosting. It excels at capturing complex relationships between features and target variables. **XGBoost** is widely used for regression tasks involving continuous outputs. Hyperparameter optimization is applied to fine-tune the model for better predictive performance.

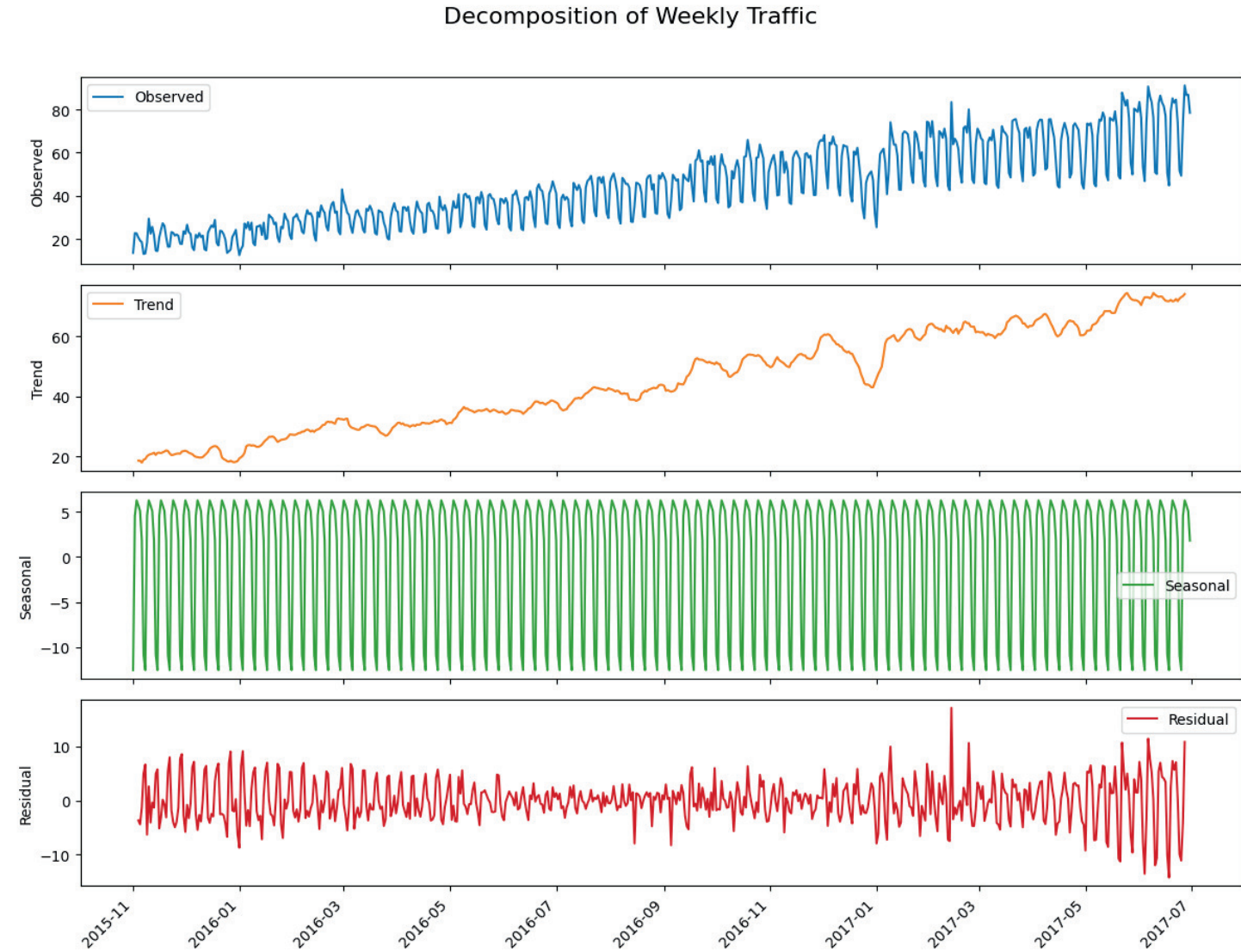
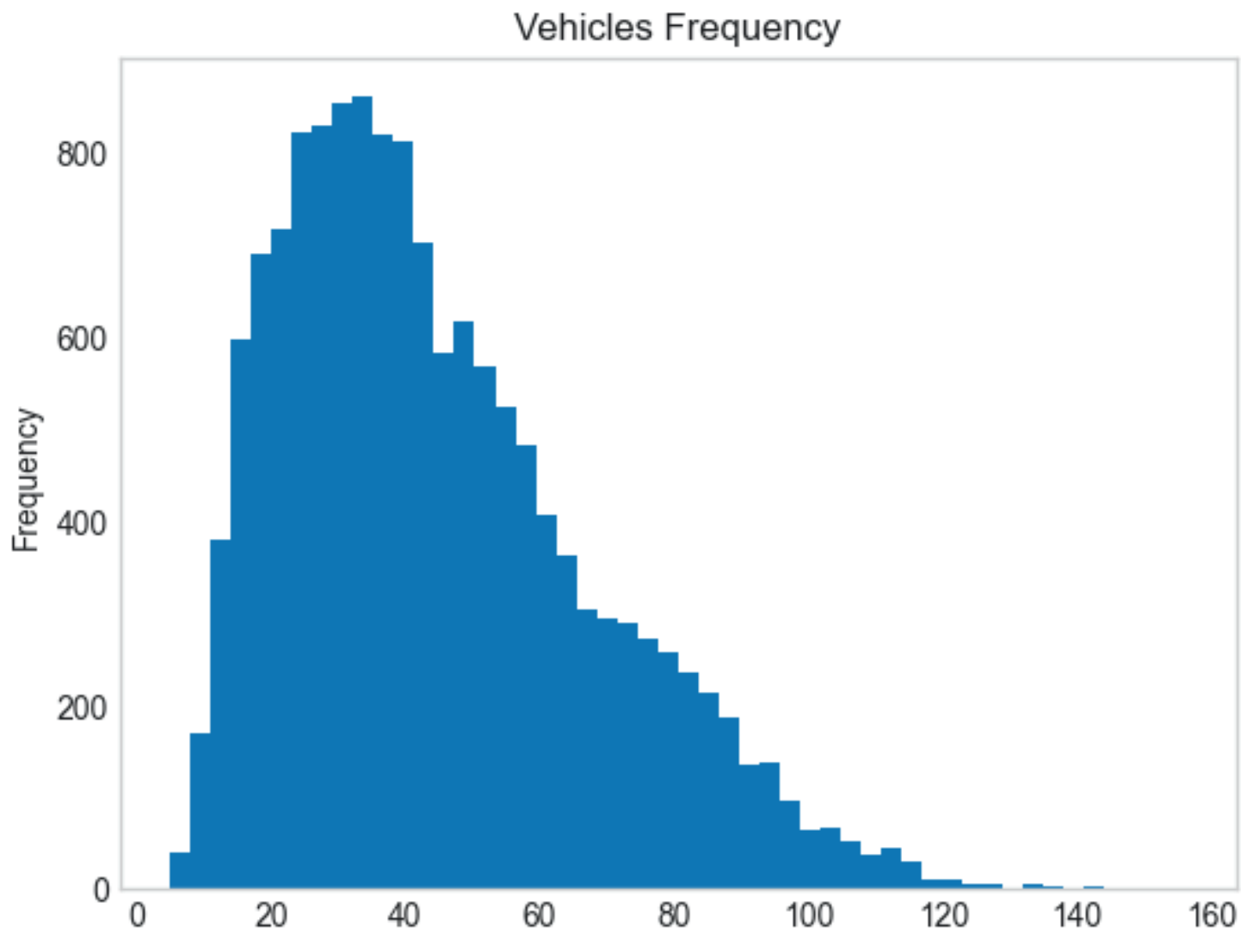


RESULT AND DISCUSSION

DATA EXPLORATION

EDA (Exploratory Data Analysis) is the process of analyzing datasets to summarize their main characteristics, often using visual methods, to uncover patterns, sport anomalies, test hypotheses, and check assumptions before formal modeling.

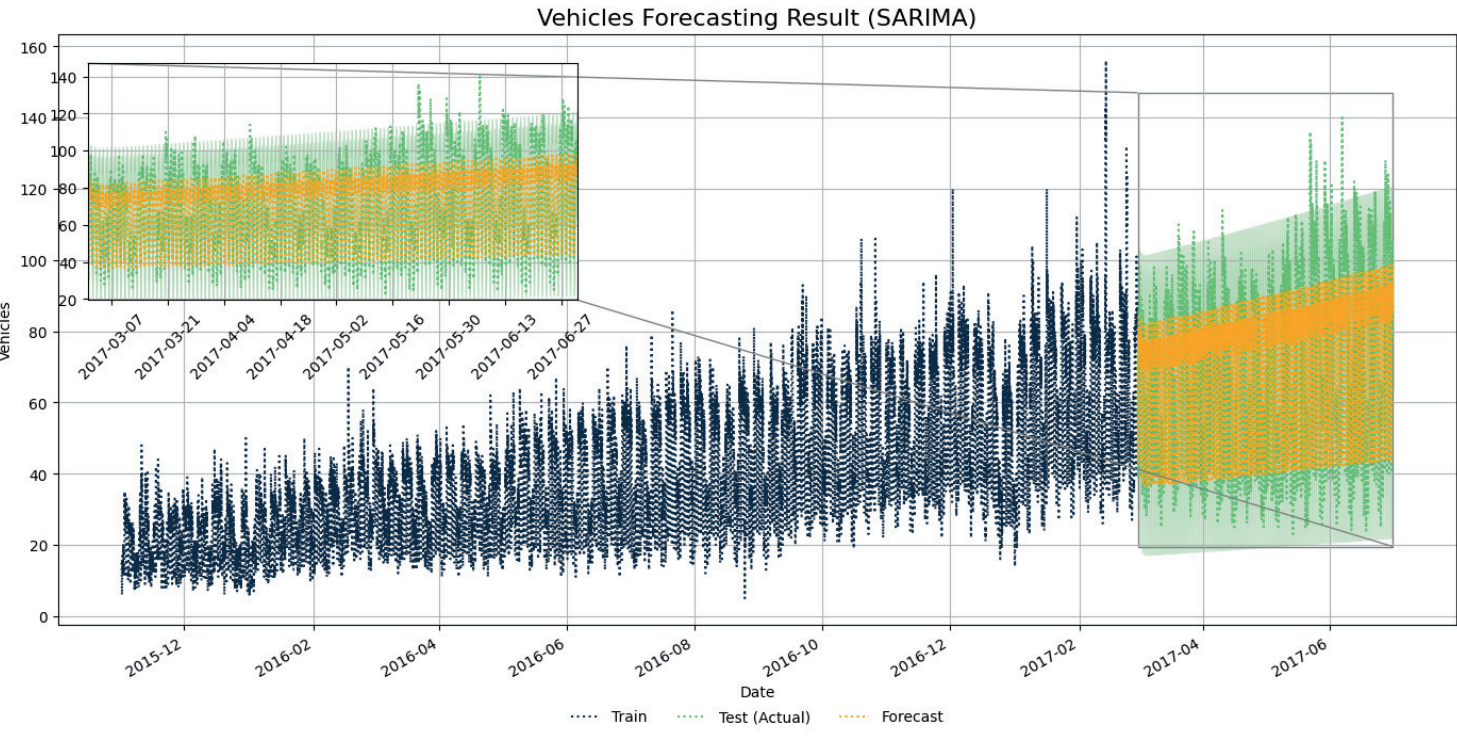
Descriptive Statistics on the Dataset	
Count	14592
Mean	45
Standard Deviation	23
Min	5
25%	27
50%	40
75%	59
Max	159



FORECASTING ON TESTING SET

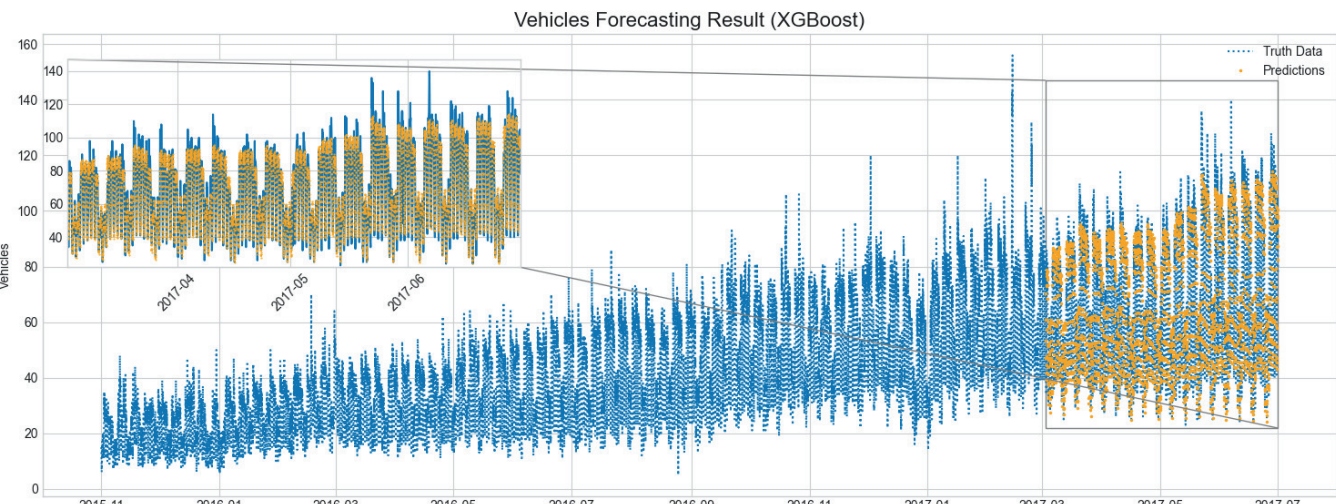
SARIMA RESULT

This is the result of **SARIMA** on both the testing sets. The graph of the predicted traffic volume is relatively close to the graph of the actual traffic volume.

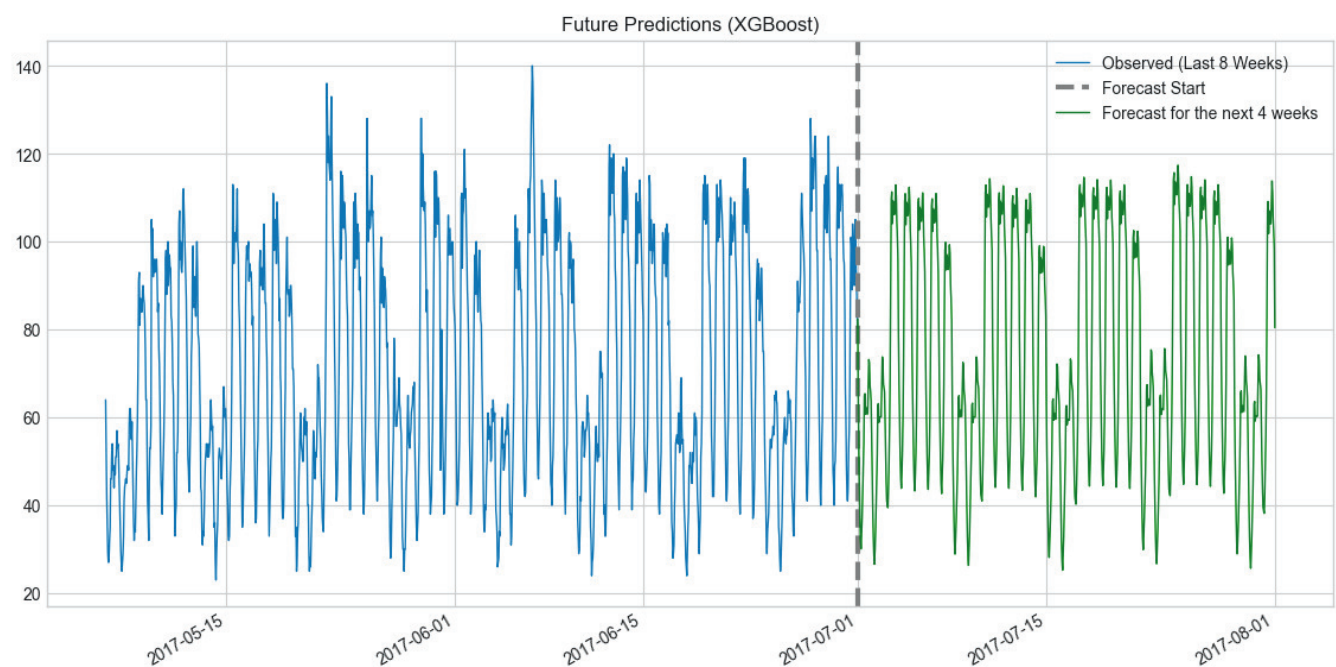
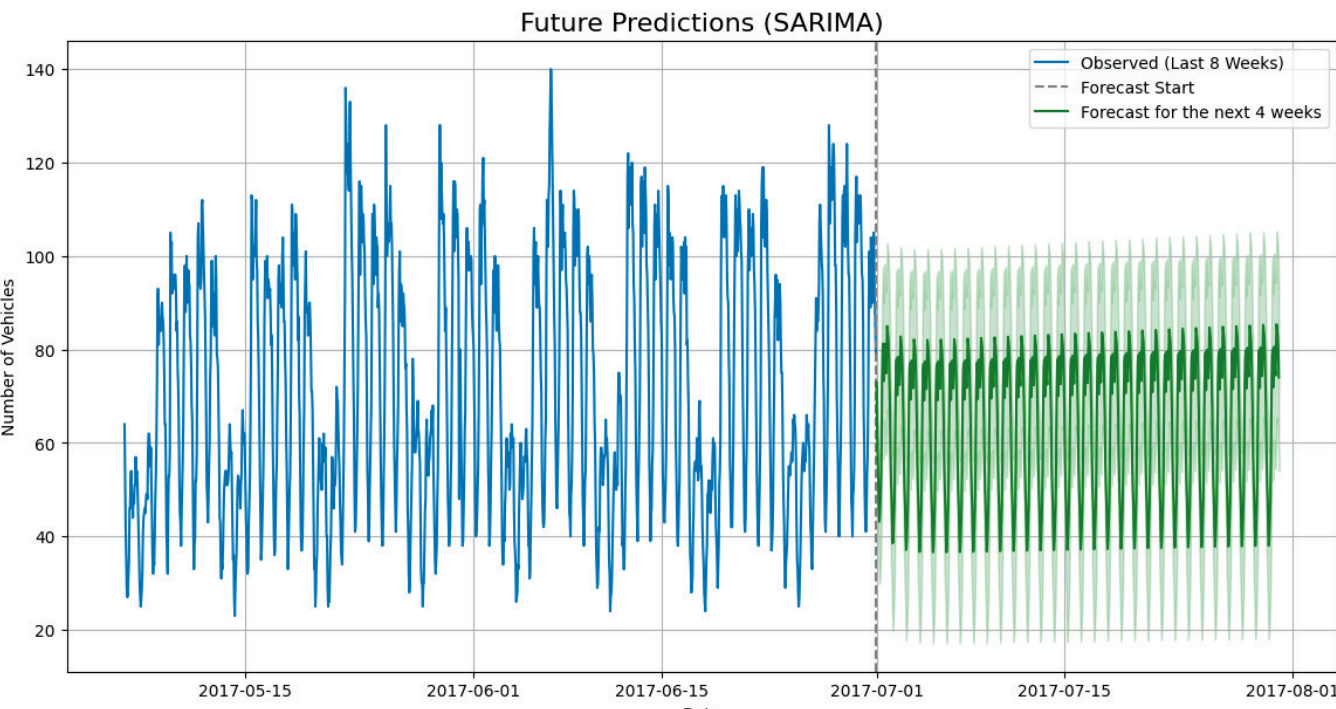


XGBoost RESULT

XGBoost on both testing sets yielded this. The predicted and actual traffic volume graphs are similar.

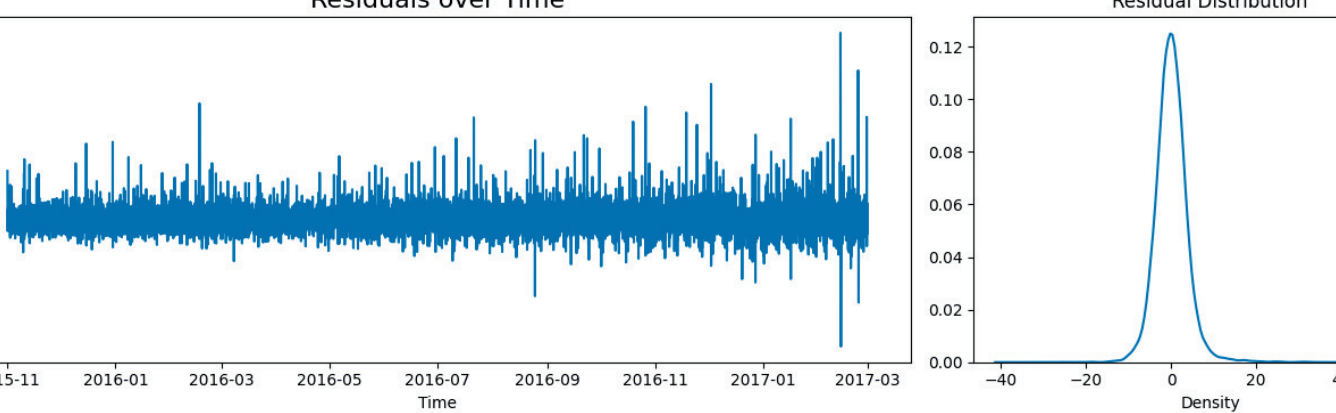


FORECAST FOR THE UPCOMING FOUR WEEKS



SARIMA MODEL RESIDUAL

When using the **SARIMA** model. It is essential to analyze the residuals to confirm an appropriate fit and that any discrepancies are randomly distributed.



MODEL PERFORMANCE COMPARISON

The results show that **XGBoost** outperforms the **SARIMA** model, providing more accurate and reliable forecasts for vehicle volume.

Model	RMSE	MAE
SARIMA (1, 0, 1)(1, 0, 1, 24)	15.4516	11.8184
XGBoost	5.1313	3.6918

CONCLUSION

This study shows that **XGBoost** predicts vehicle volume better than **SARIMA**. Its ability to handle complex patterns and incorporate features improves traffic forecasting, enabling better transportation planning and congestion management.